Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 1 (Generated practice paper - Set 13)

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
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11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 The Universal Soil Loss Equation (USLE) is A = R x K x LS x C x P. Table 1 gives factor values for a farm.

Table 1

Factor	Description	Value
R	Rainfall erosivity for the farm	110
K	Soil erodibility: loam	0.67
LS	Slope length and gradient: 120 m, 6%	1.35
C (crop type x tillage)	Beans (0.50) with spring ploughing (0.90); or wheat (0.35) with mulch tillage (0.60)	see question
P	Ploughing up and down slope 1.0; contour ploughing 0.50; strip cropping with contour ploughing 0.25	see question

Table 2

Soil erosion class	Rate of soil loss / t ha^{-1} yr^{-1}
Low	under 5
Moderate	5-10
High	10-20
Very high	20-30
Severe	over 30

0 1 . 1 Calculate the annual soil loss for beans with spring ploughing, ploughed up and down the slope. Give your answer to one decimal place.

[2 marks]

Answer _____ t/ha/yr

0 1 . 2 Calculate the annual soil loss for wheat with mulch tillage and strip cropping with contour ploughing.

[2 marks]

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Answer [ ] t/ha/yr

0 1 . 3 The natural rate of soil formation on the farm is 1 t ha^{-1} yr^{-1}. Use your answers and Table 2 to evaluate the sustainability of the two farming systems.

[3 marks]

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0 1 . 4 Explain why the USLE gives only an estimate of soil loss.

[2 marks]

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0 1 . 5 Suggest which single factor in the USLE a farmer can change most easily.

[1 mark]

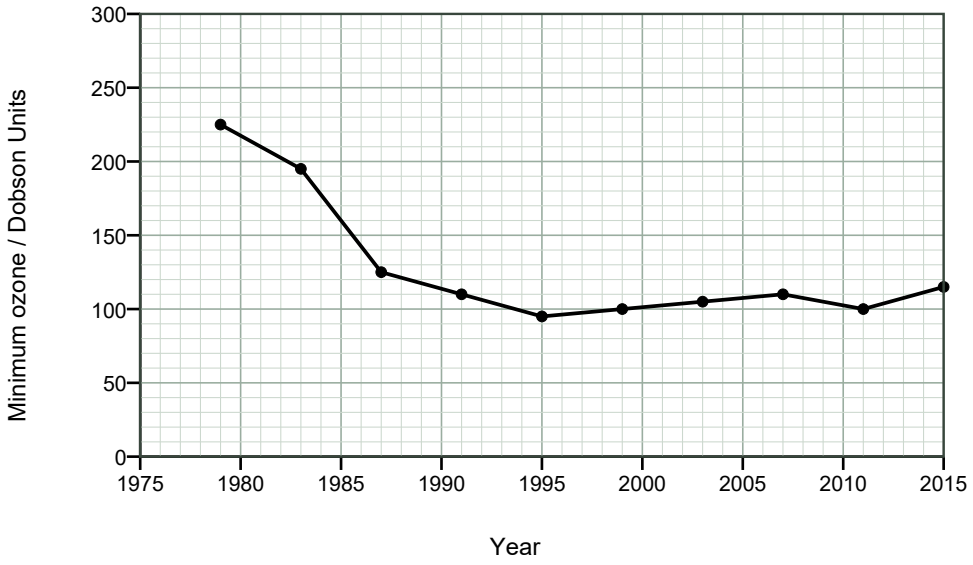
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0 2 Figure 1 shows the minimum ozone concentration recorded over Antarctica each spring between 1979 and 2015.

Figure 1



0 2 . 1 Describe the trend shown in Figure 1.

[2 marks]

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0 2 . 2 Calculate the percentage decrease in minimum ozone between 1979 and 1995.

[2 marks]

Answer ..... %

0 2 . 3 Explain why ozone depletion is most severe over Antarctica in spring.

[4 marks]

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0 2 . 4 Explain why the data in Figure 1 alone could not confirm that CFCs were the cause of the depletion, and describe how the cause was confirmed.

[2 marks]

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0 3 . 1 Explain why nitrates in drinking water can cause **blue baby syndrome** (methaemoglobinaemia).

[3 marks]

0 3 . 2 Explain why the link between nitrates in drinking water and stomach cancer is uncertain.

[2 marks]

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0 3 . 3 Describe **three** ways farmers in a Nitrate Vulnerable Zone can reduce nitrate leaching into aquifers.

[3 marks]

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0 3 . 4 Explain how phosphates are removed from sewage effluent and from lakes polluted in the past.

[2 marks]

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0 4 . 1 Describe how bioleaching is used to extract copper from low-grade ores or spoil heaps, and explain why it may be more sustainable than smelting.

[5 marks]

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0 4 . 2 Describe the process of phytomining.

[3 marks]

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0 4 . 3 Explain how polymer adsorption may provide a low-energy source of uranium in the future.

[2 marks]

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0 5 . 1 Explain how the control of emission **location** and emission **timing** can reduce the severity of pollution. Use examples.

[6 marks]

Emission location .....

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Emission timing .....

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0 5 . 2 Explain the **polluter pays** principle and suggest why it encourages pollution prevention.

[2 marks]

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0 5 . 3 Explain the **precautionary principle** as applied to the release of a new chemical.

[2 marks]

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0 6 A student estimated the population of ground beetles in a field using the Lincoln Index. 60 beetles were caught in pitfall traps, marked with a spot of paint and released. Three days later 45 beetles were caught, of which 9 were marked.

0 6 . 1 Calculate the estimated population size.

[2 marks]

Answer ____ beetles

0 6 . 2 State **four** assumptions of the Lincoln Index method.

[4 marks]

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0 6 . 3 Explain how the estimate would be affected if the paint made the beetles more likely to be eaten by birds.

[2 marks]

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0 6 . 4 Suggest why the second sample was taken after three days rather than the next day or after a month.

[2 marks]

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0 7 Mercury released into a lake from a factory was in the inorganic form. Fish in the lake were later found to contain high concentrations of methyl mercury, an organic form.

0 7 . 1 Suggest how the inorganic mercury was converted to methyl mercury.

[2 marks]

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0 7 . 2 Explain why methyl mercury is a more hazardous pollutant than inorganic mercury.

[3 marks]

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0 7 . 3 Explain why the health risk from eating fish from the lake may remain for many years after the discharges stop.

[2 marks]

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0 7 . 4 Suggest **three** factors that should be included in a Critical Pathway Analysis of a proposed mercury discharge.

[3 marks]

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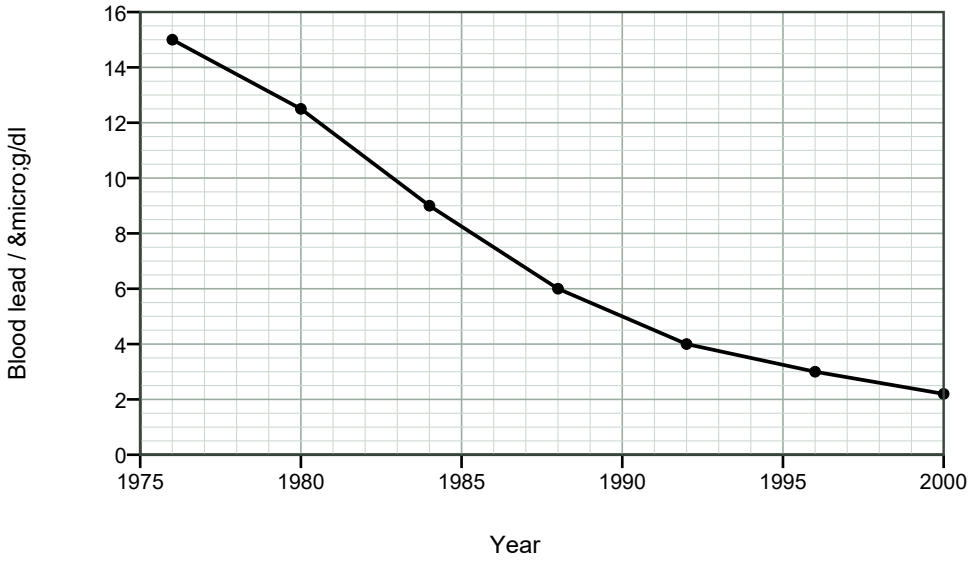
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0 8 Figure 1 shows the mean concentration of lead in the blood of children in a large city between 1976 and 2000, and the year in which lead additives in petrol began to be phased out.

Figure 1



0 8 . 1 Calculate the percentage decrease in mean blood lead between 1976 and 2000.

[2 marks]

Answer ..... %

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0 8 . 2 Explain why tetraethyl lead was added to petrol and why this caused a public health threat.

[3 marks]

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0 8 . 3 Suggest **two** reasons, other than petrol additives, why blood lead concentrations continued to fall after 1990.

[2 marks]

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0 8 . 4 Explain how adding small amounts of phosphoric acid to a water supply reduces lead in drinking water.

[2 marks]

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0 8 . 5 Explain why discarded lead fishing weights and shotgun pellets can poison waterfowl.

[1 mark]

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0 9 . 1 Which of the following properties would make a pesticide **least** likely to biomagnify? Tick one box.

[1 mark]

A High liposolubility and high persistence ☐

B Low liposolubility and low persistence ☐

C High liposolubility and low persistence ☐

D Low liposolubility and high persistence ☐

0 9 . 2 Explain the difference between **bioaccumulation** and **biomagnification**.

[2 marks]

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0 9 . 3 Explain why pesticides are described as an unusual type of pollutant.

[2 marks]

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1 0 . 1 Explain why the development of energy storage technologies is essential if renewable energy resources are to provide most of a country's electricity.

[9 marks]

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1 0 . 2 State what is meant by **peak shaving**.

[1 mark]

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Write an essay on one of the following topics.

1 1 . 1 Discuss how knowledge of the properties of pesticides can be used to reduce their environmental impacts while maintaining food production.

[25 marks]

OR

1 1 . 2 Discuss how knowledge of the properties of pollutants can be used to reduce the severity of their impact on the environment.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS